

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.330034453562399, median 0.28149819687378286, std: 0.2236441642729055
Gyroscope error (imu0): mean 1.088771020505368, median 0.8714884449039665, std: 0.8902782412886165
Accelerometer error (imu0): mean 0.5151848724176596, median 0.437466462851411, std: 0.8708773840448254

Residuals

Reprojection error (cam0) [px]: mean 0.330034453562399, median 0.28149819687378286, std: 0.2236441642729055
Gyroscope error (imu0) [rad/s]: mean 0.080321429801784, median 0.06429193708510791, std: 0.06567810854161761
Accelerometer error (imu0) [m/s²]: mean 0.00972461938927971, median 0.008257608238456917, std: 0.016438664153363827

Transformation (cam0):

T_ci: (imu0 to cam0):

[[0.00143535 -0.9996217 -0.0274663 0.01027165]
[-0.99999271 -0.00133764 -0.00357536 -0.00646529]
[0.00353727 0.02747123 -0.99961634 -0.0257764]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[0.00143535 -0.99999271 0.00353727 -0.00638881]
[-0.9996217 -0.00133764 0.02747123 0.01096722]
[-0.0274663 -0.00357536 -0.99961634 -0.0255075]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.1219170237917244

Gravity vector in target coords: [m/s²]

[-0.21846683 -9.80401437 -0.04469101]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [543.7118550256475, 543.8969677210899]

Principal point: [338.27510774897553, 200.12046609071774]

Distortion model: equidistant

Distortion coefficients: [0.3736158484120638, 0.4917026512938491, -1.2722127894904778, 0.9892679714070943]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.021 [m]

Spacing 0.0063 [m]

IMU configuration

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IMU0:

Model: scale-misalignment

Update rate: 100.0

Accelerometer:

Noise density: 0.0018875980080012855

Noise density (discrete): 0.018875980080012855

Random walk: 2.553493980803048e-05

Gyroscope:

Noise density: 0.007377256400937428

Noise density (discrete): 0.07377256400937428

Random walk: 3.250380706253323e-05

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

[0. 1. 0. 0.]

[0. 0. 1. 0.]

[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

Gyroscope:

M:

[[0.94499723 0. 0.]

[0.00456959 0.96665036 0.]

[-0.04313862 0.04449462 0.99311559]]

A [(rad/s)/(m/s²)]:

[[0.00103668 -0.00180641 -0.00095881]

[0.00684194 -0.00186347 -0.00647335]

[0.00394249 0.00159014 0.0015275]]

C_gyro_i:

[[0.99946771 0.00036475 -0.03262151]

[0.00049534 0.99965255 0.02635391]

[0.03261979 -0.02635604 0.99912027]]

Accelerometer:

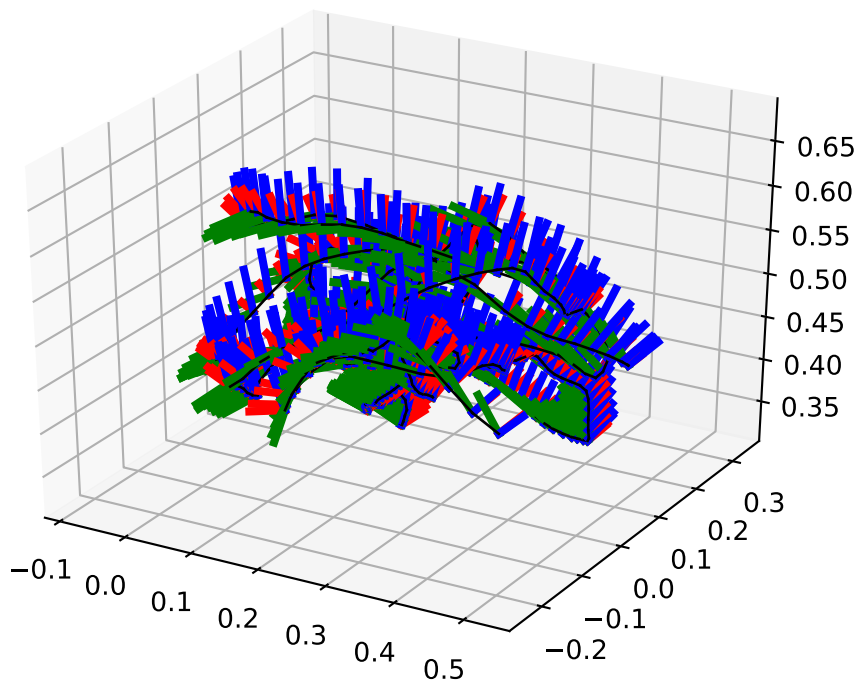
M:

[[1.00032056 0. 0.]

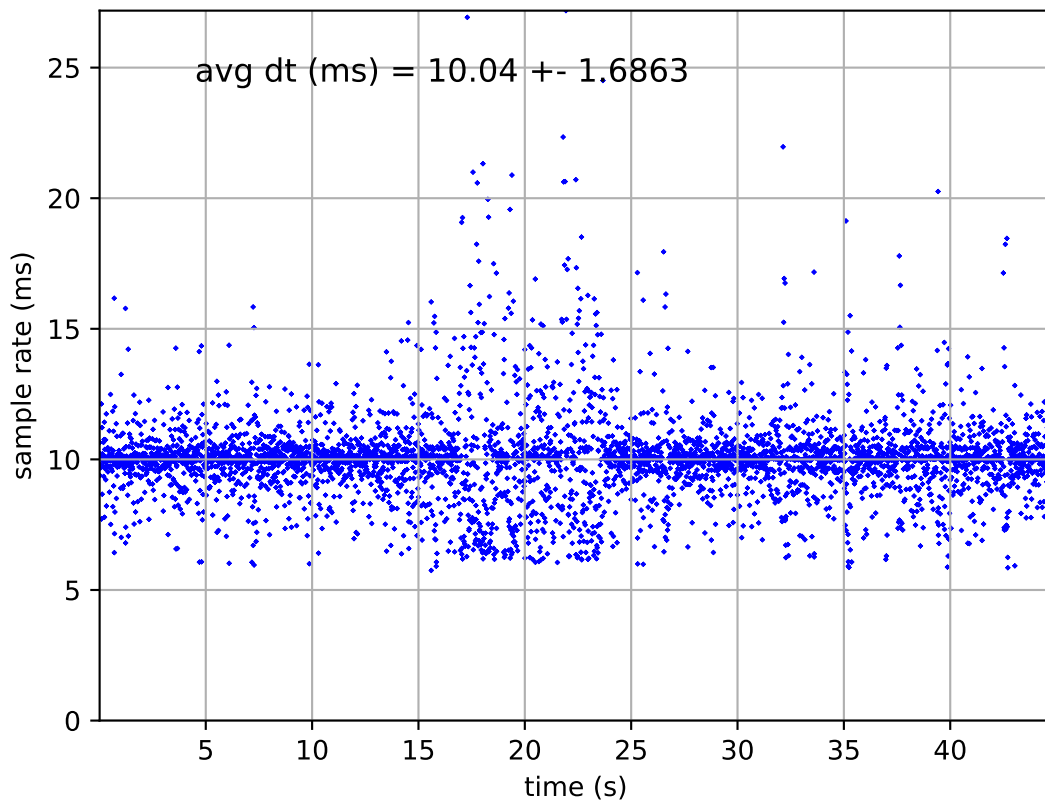
[-0.00003534 0.99564465 0.]

[0.00091423 -0.00464574 0.9723505]]

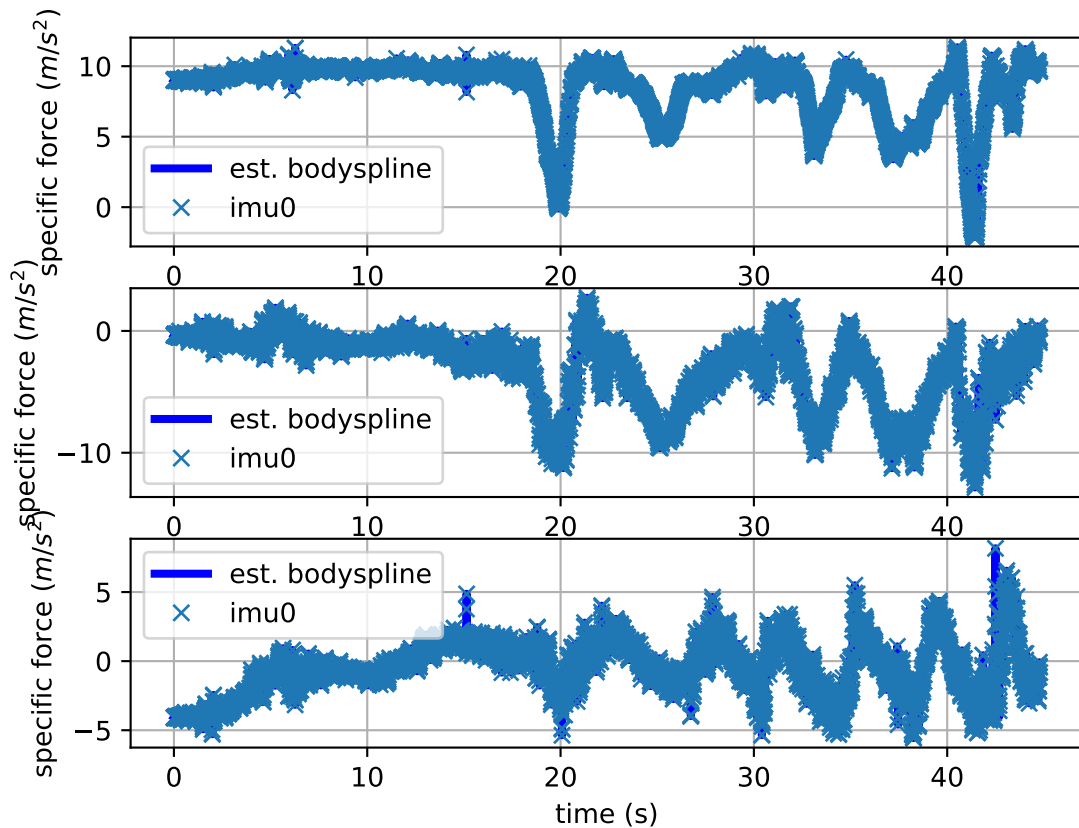
imu0: estimated poses



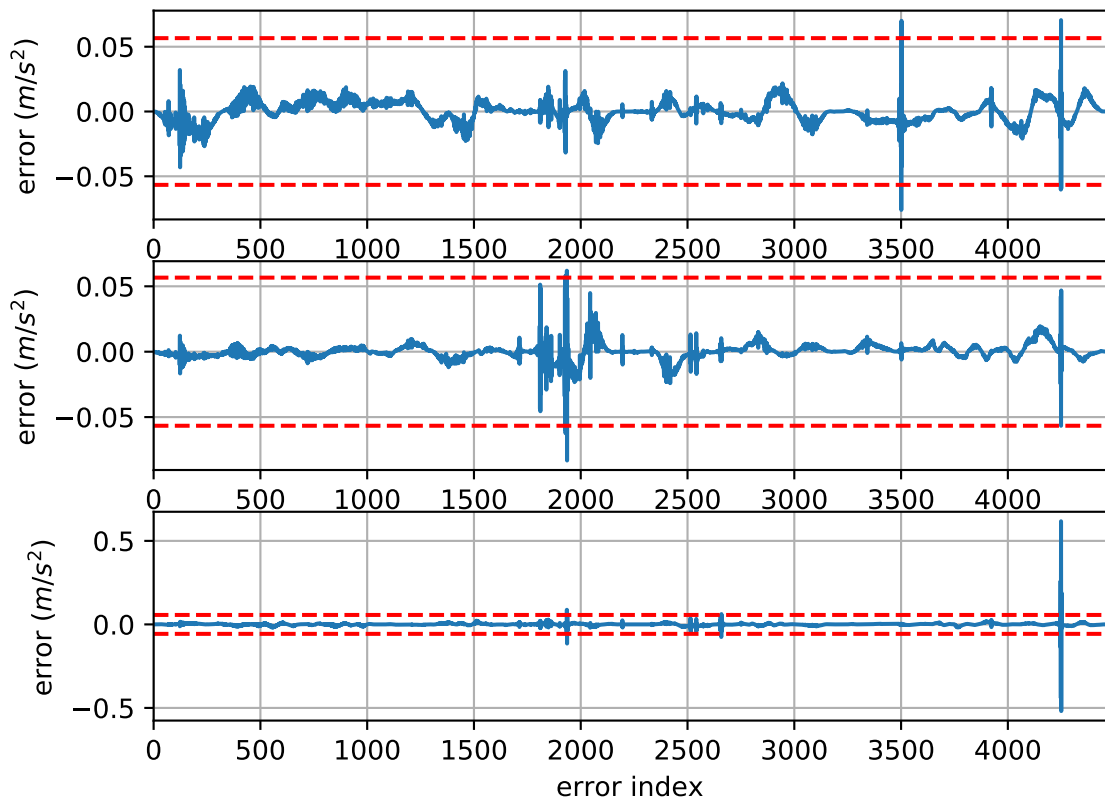
imu0: sample inertial rate



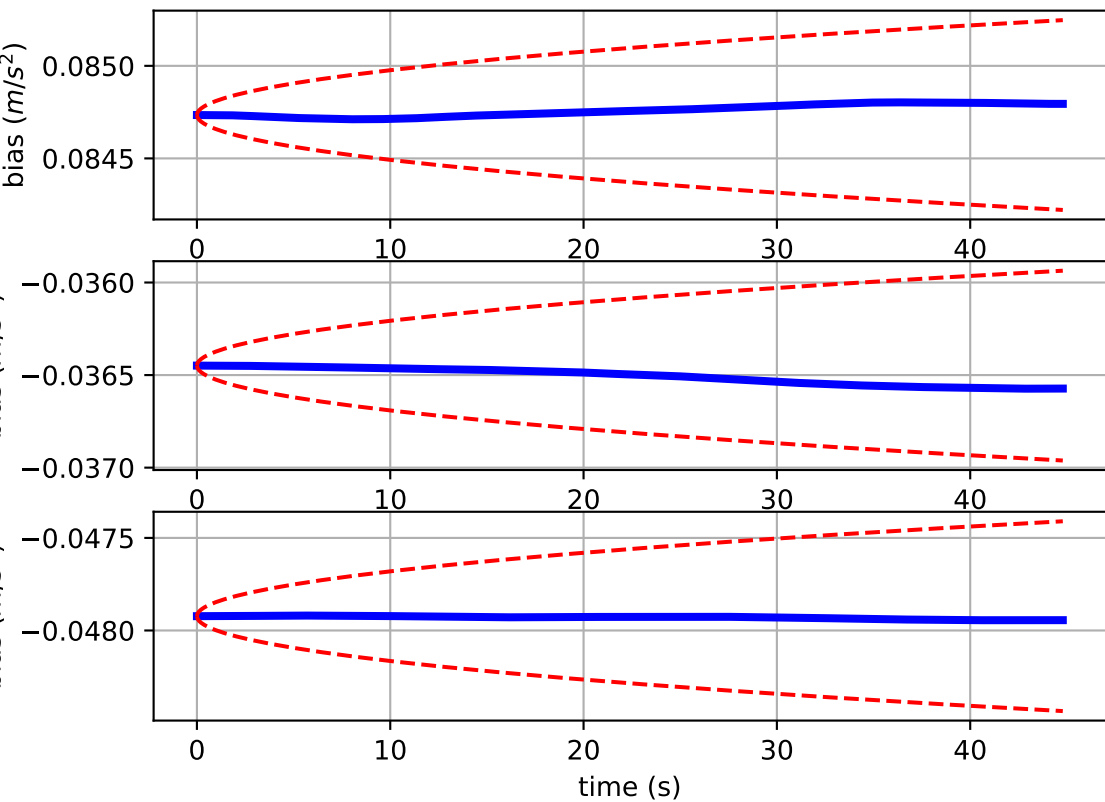
Comparison of predicted and measured specific force (imu0 frame)



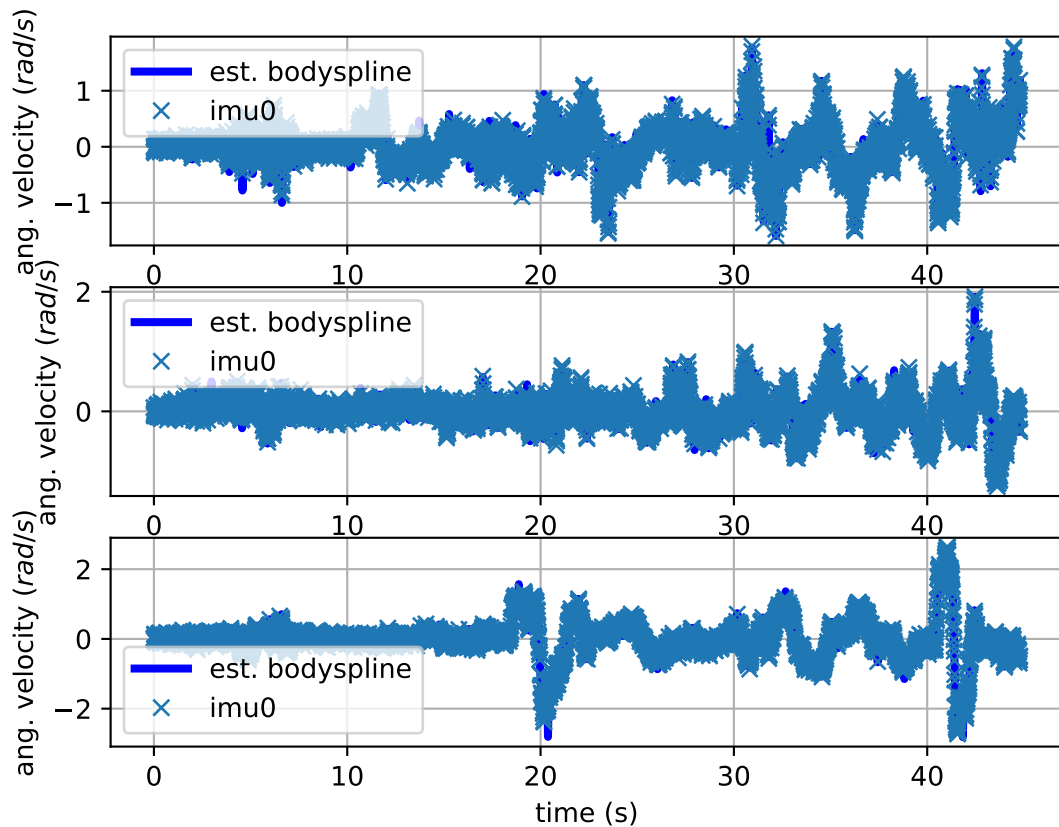
imu0: acceleration error



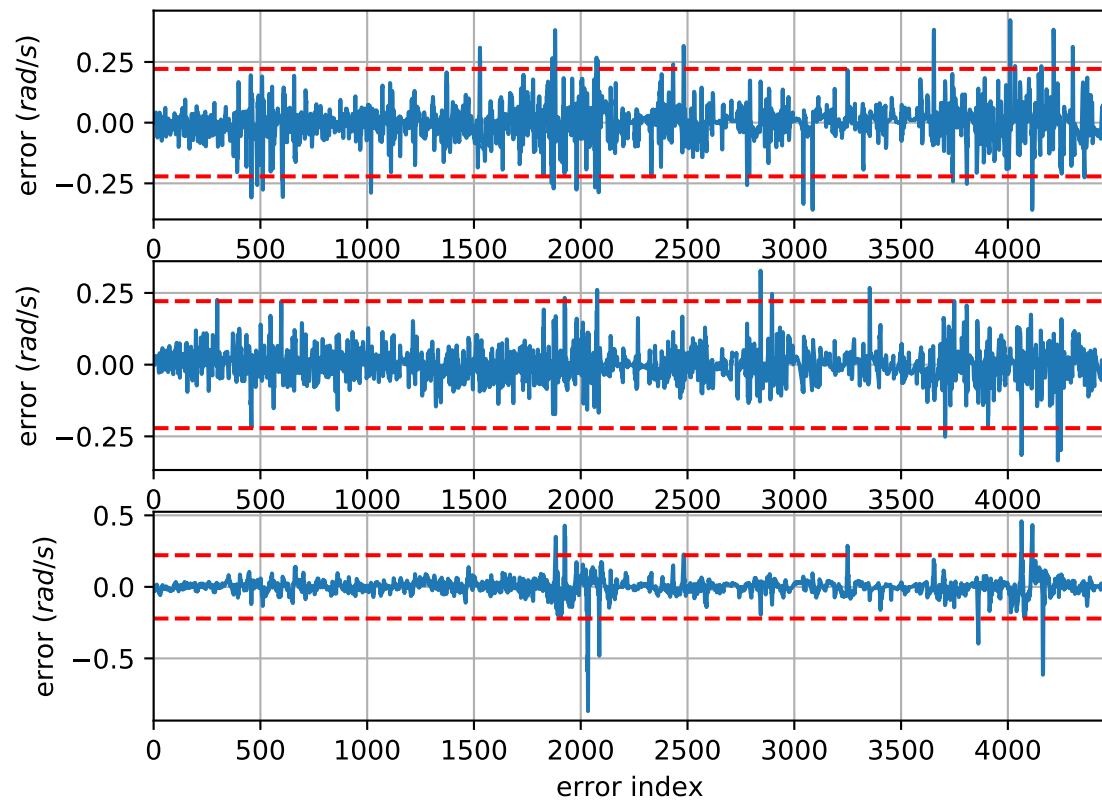
imu0: estimated accelerometer bias (imu frame)



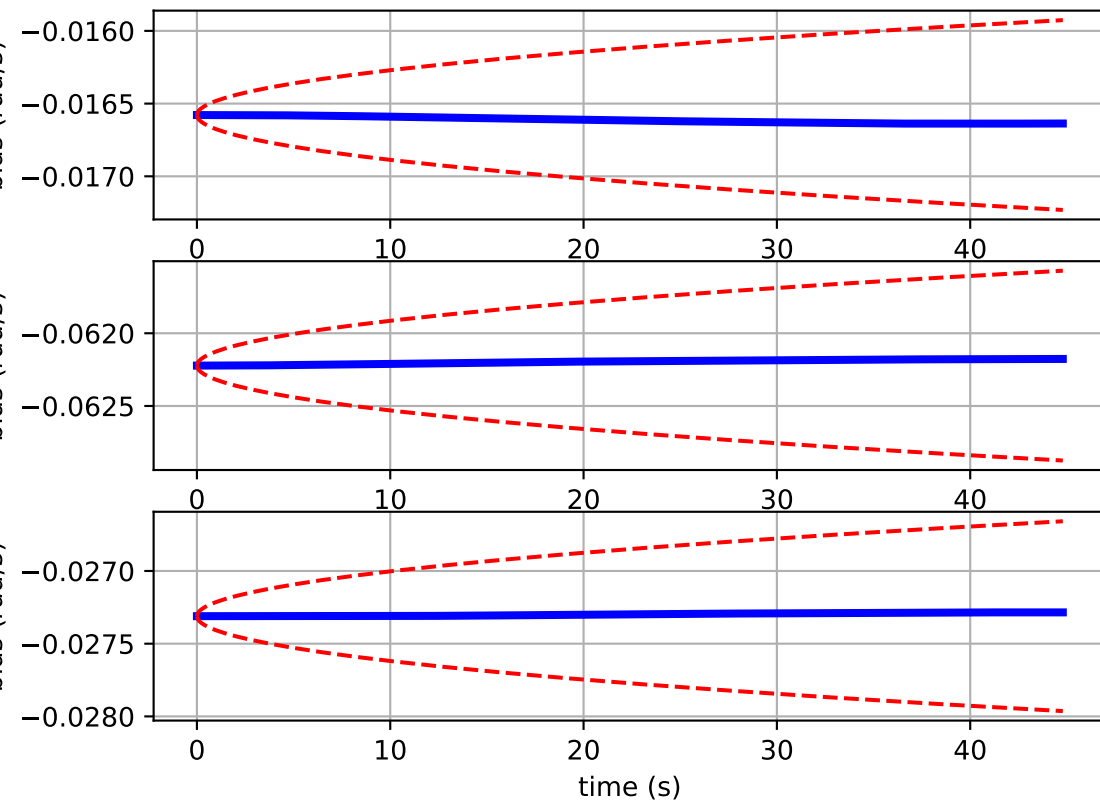
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

